

objectives for today:

- precise description of the algo.

issues I had with it:

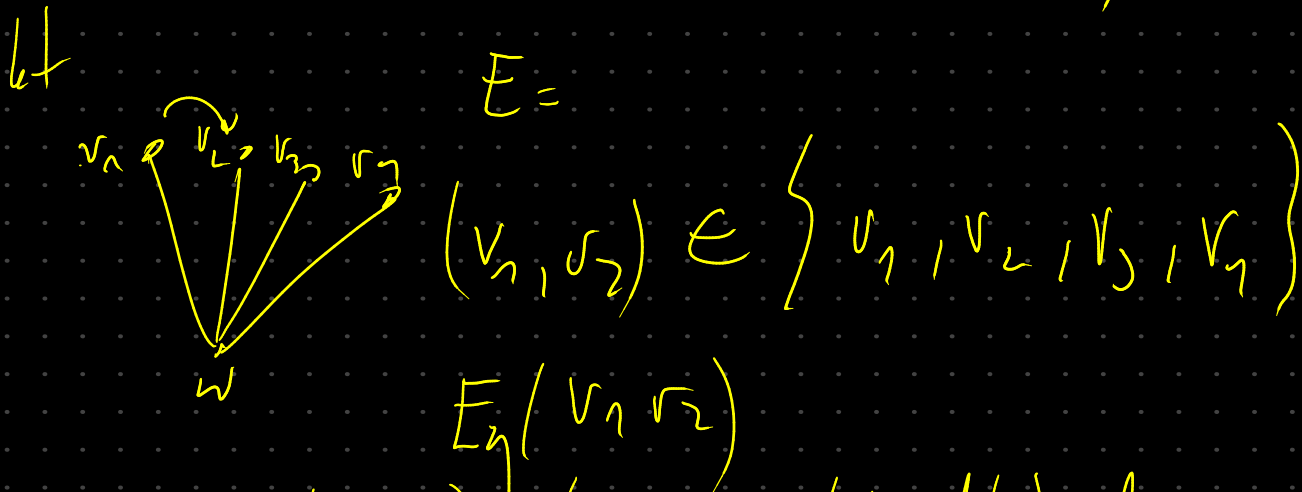
synchronizing equivl classes  
(define eq classes that are symmetric for the mapping)

\* Idem:

Input:  $h$  level planar graph  
Eq classes (orbit)

$\forall u, v \in \mathcal{L}(\alpha/\beta): u < v \iff \alpha < \beta$

$$Eq(\alpha/\beta) = \left\{ u, v \in V(h), l(u) = l(v) \text{ s.t. } u < v \iff \alpha < \beta \right\}$$

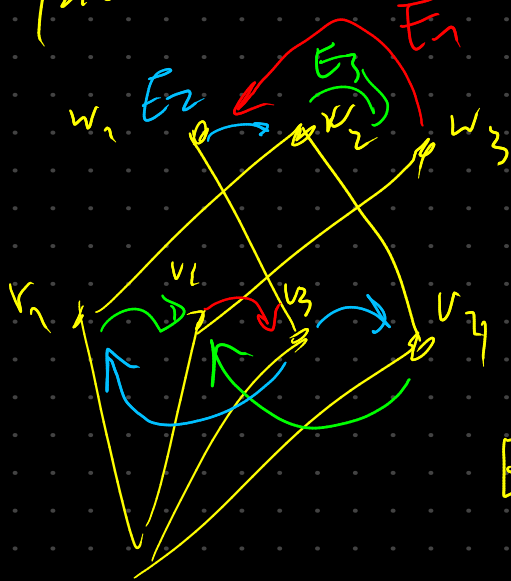


$$E_q(v_1, v_2) = \left\{ u, v \in V(h), l(u) = l(v) \text{ s.t. } u < v \iff (\alpha < \beta) \text{ or } (\beta < \alpha) \right\}$$

$$E_{q_{abs}}(v_1 v_2) = E_q(v_1 v_2) \cup \bar{E}_q(v_2 v_1)$$

$$E_{q_{abs}}(v_1 v_2) = 1 \Rightarrow \begin{cases} E_q(v_1 v_2) = 1 \\ E_q(v_2 v_1) = 0 \end{cases}$$

$$E_{q_{abs}}(v_1, v_2) = E_{q_{abs}}(v_3, v_4)$$



$$E_1(v_1 v_2) = \{ \overset{(v_1, v_2)}{(v_2 v_3), (v_4 v_2)} \}$$

$$E_q(v_2, v_1) = \{ (w_3 w_2), (v_2 v_4) \}$$

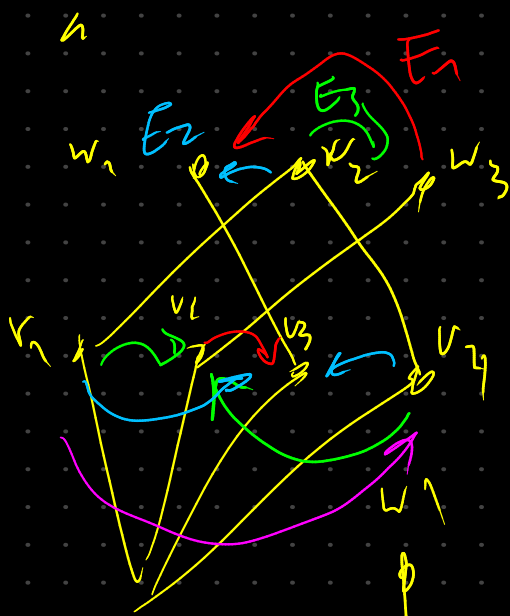
$$E_{q_{abs}}(v_1, v_2) = \left\{ \begin{matrix} \{v_1, v_2\} \\ \{w_2, w_3\}, \{v_2, v_4\} \end{matrix} \right\}$$

$$E_{q_{abs}}(v_2, v_3) = \{ \{w_1, w_3\} \}$$

$$E_{q_{abs}}(v_1, v_3) = \{ \{v_3, v_4\}, \{w_1, w_2\} \}$$

non cyclic relation for distinct  $E_{q_{abs}}$  classes

|      |   |   |
|------|---|---|
| blue | ↪ | R |
| ↱    | ↱ | ↱ |



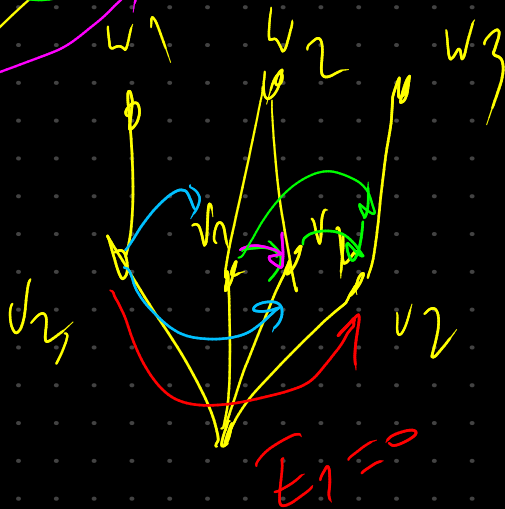
$$\bar{E}_2 = 1$$

$$\bar{E}_1 = 1$$

$$\bar{E}_3 = 0$$

$$E_1(v_1, v_2) = \{v_1 v_2, v_1 v_3\} \in [v_2, v_3]$$

$$E(v_1, v_2) = \{E(v_2, v_1)\}$$



$$\nexists v_1 v_2 \in E(d|\beta)$$

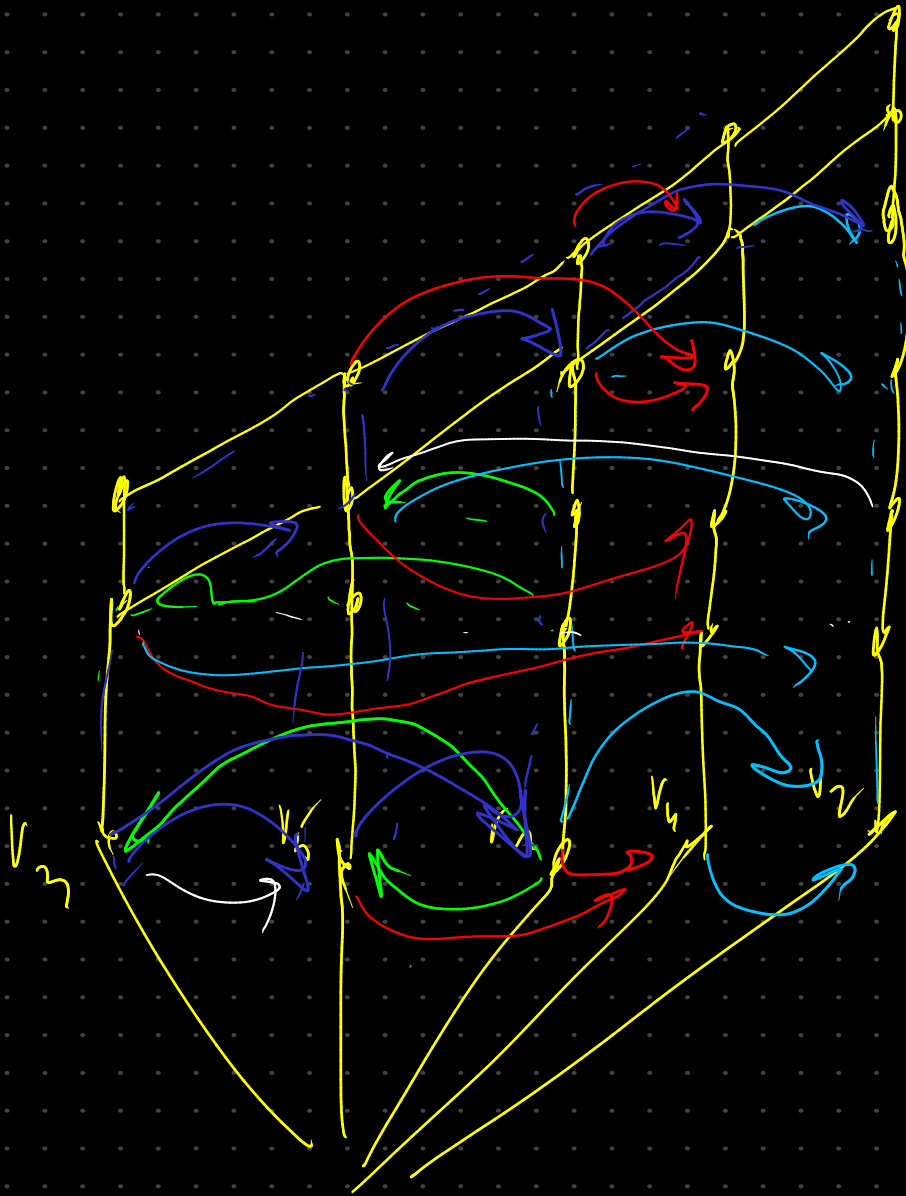
$$v_2 v_1 \in E(\beta_\alpha)$$

$$E_1(v_1, v_3) = [v_2, v_1, v_3, v_1]$$

$$v_1 \in E(d|\beta) \Leftrightarrow v_1 \in E(\beta_\alpha)$$

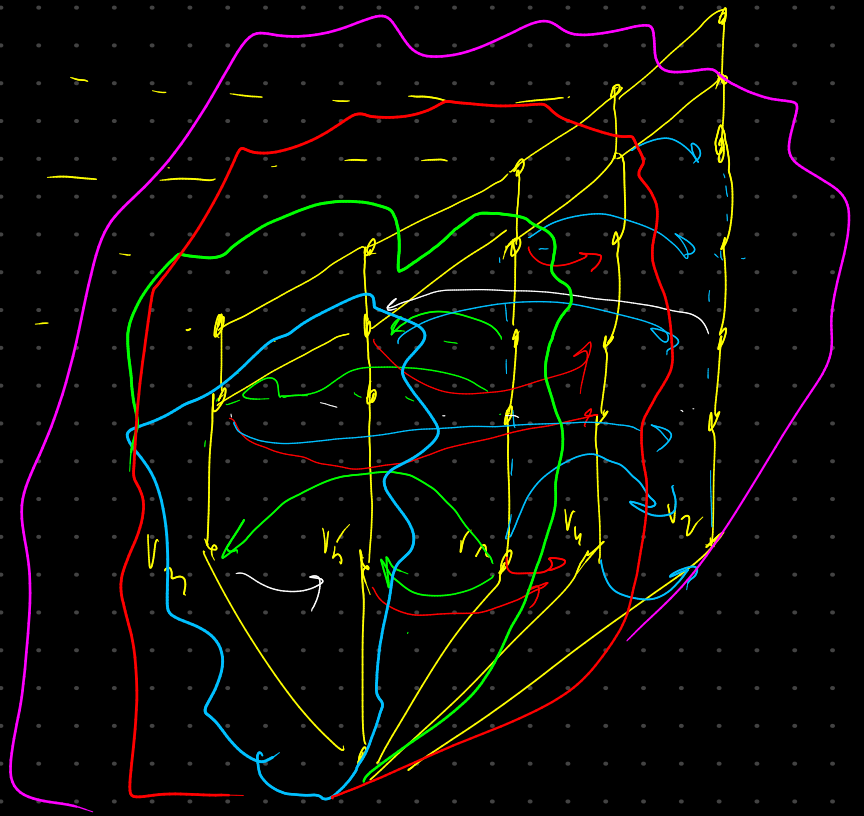
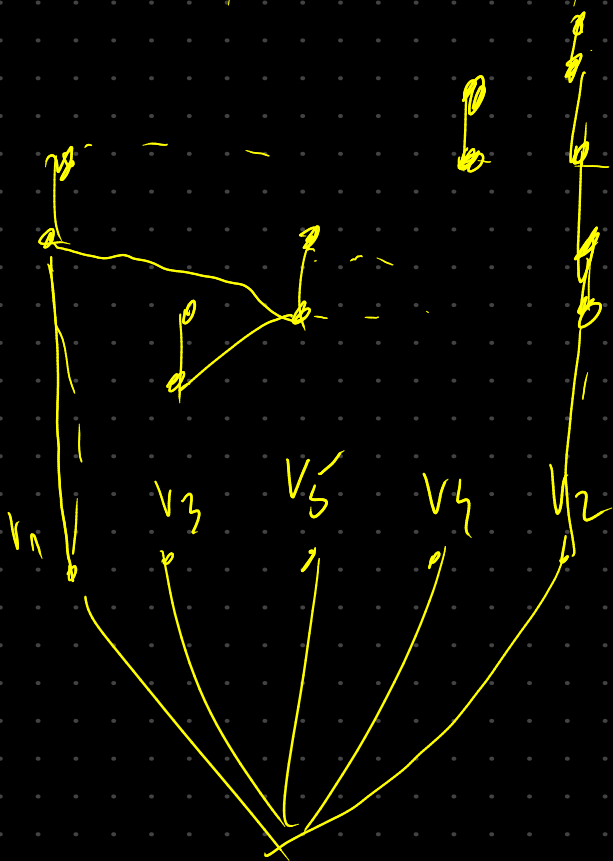
$$E_1(v_1, v_2) = \begin{matrix} E(v_1, v_2) \\ E(v_2, v_1) \\ E(v_2, v_3) \\ E(v_3, v_2) \end{matrix} = \begin{matrix} E(v_1, v_2) \\ E(v_2, v_1) \\ E(v_2, v_3) \\ E(v_3, v_2) \end{matrix}$$

$$E(v_1, v_2) = E(v_1, v_3) = E(v_1, v_4) = E(v_2, v_3)$$

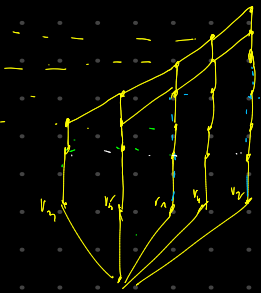


we will have one equivalent class cluster.

$$E(v_1, v_2) = E(v_1, v_3) = E(v_1, v_4) = E(v_1, v_5)$$



how to define this?

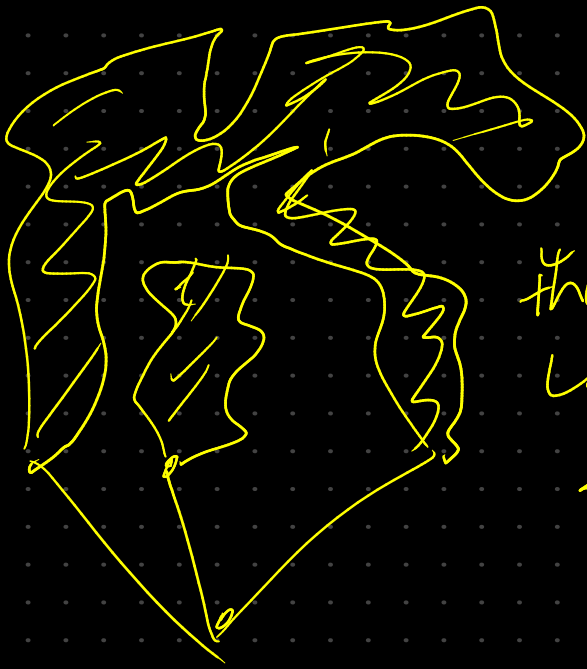


not part  
of the core

that  
to an  
odd group  
in H-T

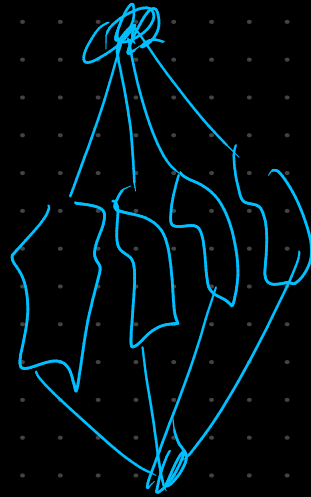


with v0 is connected



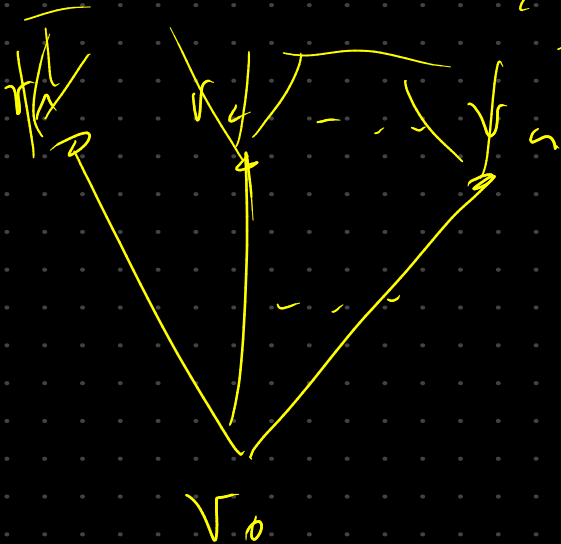
theorem 3.1  
 case  
 2. a)

and  
 $h = \{v_0, h\}$   
 is still  
 connected  
 so we are  
 not in the  
 case of



but how to  
 represent the  
 $\gamma$  class?

suppose



we want to find  
 $\gamma$  classes that depend  
 on each other  
 $\gamma\text{-classes\_cluster} = \{ \}$

if  $(v_i, v_j) \notin \gamma\text{-L\_C}(k)$   
 and  
 $(v_j, v_i) \notin \gamma\text{-L\_C}(k)$

$\gamma\text{-L\_C\_cluster.add}(\gamma(v_i, v_j))$

$$eq\_c\_c(i) = eq\_c\_c(j) \quad \forall 0 \leq i < j \leq n$$

definitions

— equivalent class

— equivalent class cluster <sup>maybe better naming</sup>

— troublesome vertex (for later)

— eq class

$$Eq(\alpha/\beta) = \{ v < w \Leftrightarrow \alpha < \beta \mid v, w \in V(n) \\ l(v) = l(w) \}$$

with  $\alpha/\beta$  represent the ordering

<sup>with  $v < w$  and  $v.id < w.id$</sup>

$$V = \{ uv \mid u, w \in V, u \neq w, l(u) \neq l(w) \}$$

$$\forall uv \in V \quad Eq(uv) = \{ uv \Leftrightarrow vx \}$$

$$uv \Leftrightarrow vx$$

$$\forall uv \Leftrightarrow \neg(vx)$$

$$vx \in V \text{ and}$$

$$\begin{cases} u \cdot i \subset w \cdot i \\ v \cdot i \subset x \cdot i \end{cases}$$

(this trace becomes  
we have consistency)

$$(u, v), (w, x) \in E$$

independent

$$\forall uv \in V \quad Eq_c(uv) = \left\{ \begin{array}{l} uv \Leftrightarrow vx \vee \\ uv \Leftrightarrow \neg vx \end{array} \right\}$$

$$Eq_c(uv) = Eq(uv) \cup Eq(wu)$$

We exclude pair that are identical  $\checkmark$   
(need more context)

$$\forall uv, vx \in V \quad l(u) = l(v)$$

$$Eq_c(uv) = Eq_c(vx)$$

$$Eq_c(uv) = \bigcup_{vx \in \text{path}(v)} Eq_c(vx)$$



$$\forall uv, v \in V \left\{ \begin{array}{l} u.id < v.id \\ v.id < u.id \end{array} \right.$$

$$v \in Eq(uv) \vee Eq(vu) \Rightarrow Eq$$

$$Eq_{cluster}(uv) = \{ v \in Eq(uv) \vee v \in Eq(vu) \}$$

$$Eq_{cluster}(uv) = \{ v \mid v \in Eq(uv) \} \cup \{ v \mid v \in Eq(vu) \}$$

$$C = \{ Eq(uv) \mid \forall u, v \in V \quad v \in Eq(uv) \}$$

$$\forall uv \in E \rightarrow u.id < v.id$$

$$Eq(uv) = Eq(v, u)$$

$$Eq(uv) = \bigcup_{u'v' \in E(uv)} Eq(u'v')$$

$$\cup \{ v \mid v \in Eq(uv), v \in Eq(uv) \}$$

we give an order of vertices for each level  
 since we have consistency done satisfied  
 with current eq that we have

look for adjacent vertices

$$\text{Adj}(l, v) = \left\{ w \mid w \in V(h), (v, w) \in E(h) \right. \\ \left. l(w) = l - l(v) + 1 \right\}$$

$$\forall h \in \{1, \dots, k\} \quad E^{-1}(h) \mid (v, w) \in E(h), \\ l(v) = h - 1$$

$$\forall v \in \{V(h) \mid l(v) = h-1\}$$

$$\mathcal{V}_{\text{adj}}(v, h) = \{vw \mid w, w \in \text{adj}(v, h), w \cdot \text{id} \neq \text{id}\}$$

$$\forall vw \in \mathcal{V}_{\text{adj}}(v, h)$$

$$E_{\gamma}(vw) = \bigsqcup_{\substack{u'u' \\ \in \mathcal{V}_{\text{adj}}(v, h)}} E_{\gamma}(u'u')$$